

Traffic Flow & Congestion Prediction System

Short Project Plan • Beginner-Friendly Group Project

Goal: predict future traffic volume from historical traffic data and classify the expected congestion as Low, Moderate, or High.

1. Project Objective

The system will learn traffic patterns from historical data. Given information such as time, day, previous traffic, average speed, waiting time, and an optional event indicator, the model will predict the expected number of vehicles. The predicted volume will then be converted into a congestion category.

2. Main Inputs

Feature	Purpose
Hour	Traffic changes by time of day.
Day of week	Weekdays and weekends can have different patterns.
Vehicle count	Measures traffic volume in the historical record.
Average speed	Helps describe how freely traffic is moving.
Waiting time	Useful for identifying intersection congestion.
Previous traffic	Recent traffic helps predict upcoming traffic.
Event (optional)	Festivals/special events can change normal traffic.

3. Prediction Flow

Historical Traffic Data → Pandas Data Cleaning → Feature Preparation → XGBoost → Predicted Vehicle Count → Congestion Classification

Example

Input: 9 AM, Monday, previous traffic = 850, average speed = 18 km/h, waiting time = 2 minutes, event = Yes.

Output: predicted traffic = approximately 1050 vehicles → **High Congestion**.

4. Congestion Classification

The ML model's main target is **traffic volume**, not a hard-coded congestion label. After prediction, suitable traffic-volume thresholds for the selected road/dataset can be used to label the result as Low, Moderate, or High. Thresholds should be justified from the dataset rather than presented as universal traffic rules.

5. Recommended Scope

Keep in Basic Version	Future / Optional
Historical traffic data	Real-time CCTV vehicle detection
XGBoost prediction	LSTM
Vehicle count, speed, waiting time	GNN / spatial modelling
Hour, day, previous traffic	Advanced weather features
Simple event indicator	Complex probability/Bayesian model

Technology Stack & System Design

6. Technology Stack

Technology	Use in Project
Python	Main programming language.
Pandas	Load, inspect, clean, and prepare traffic data.
Scikit-learn	Train/test splitting and evaluation utilities.
XGBoost	Main machine-learning model for traffic-volume prediction.
Matplotlib	Simple charts for data analysis and model results.
Jupyter Notebook	Beginner-friendly environment for ML development.
PostgreSQL	Store historical traffic and application data.
FastAPI	Backend API; receives inputs and returns predictions.
React	Simple frontend/dashboard for users.
Mapbox	Optional map showing roads/intersections and congestion.

7. Final System Architecture

Traffic Dataset



Pandas — cleaning & feature preparation



XGBoost — traffic-volume prediction



Predicted vehicle count



Low / Moderate / High congestion



FastAPI backend ↔ PostgreSQL database



React dashboard → Mapbox visualization

8. Development Plan

1. Create a small traffic dataset and understand every column.
2. Load and clean the dataset with Pandas.
3. Split data into training and testing sets.
4. Train the first XGBoost model.
5. Evaluate with MAE, RMSE, and R^2 .
6. Improve features using time, previous traffic, speed, waiting time, and events.
7. Save the trained model.
8. Add PostgreSQL for storing data.
9. Build a FastAPI /predict endpoint.
10. Build a simple React dashboard.
11. Add Mapbox visualization if time permits.
12. Keep CCTV, LSTM, GNN, and advanced weather as future enhancements.

9. Why This Project Is Suitable

It is small enough to build and understand as a student group, but it still demonstrates the complete ML application pipeline: **data → preprocessing → machine learning → evaluation → backend API → database → frontend → visualization**. The project can be expanded later without making the basic version unnecessarily complicated.

10. One-Sentence Viva Explanation

"Our system uses historical traffic data and factors such as time, vehicle count, average speed, waiting time, previous traffic, and special events to predict traffic volume using XGBoost and classify the expected congestion as Low, Moderate, or High."